

Performance Optimization Summary

Overview

To improve dashboard efficiency and responsiveness, several performance optimization techniques were implemented during the development of the Power BI Sales Performance Dashboard.

Optimization Techniques Implemented

Data Model Optimization

- Implemented a Star Schema model for efficient relationships.
- Reduced unnecessary columns and duplicate data.
- Maintained proper one-to-many relationships between fact and dimension tables.

Optimized DAX Calculations

Reusable DAX measures were created instead of repeated calculations to improve maintainability and processing efficiency. Functions such as `CALCULATE()`, `DIVIDE()`, and `VAR` were used to optimize calculations and improve readability.

Time Intelligence Optimization

A dedicated Calendar table was created to support accurate and efficient time intelligence analysis, including:

- Growth %
- YoY Growth
- Monthly Trends
- Moving Average calculations

Visual Optimization

Dashboard visuals were optimized by reducing unnecessary visual clutter and maintaining a clean layout structure. Interactive slicers, bookmarks, and filter panels were implemented to improve usability without affecting dashboard performance.

Optimized Interactivity

- Implemented slicers for dynamic filtering.
- Added bookmarks for reset functionality.
- Reduced unnecessary cross-filter interactions.

Interactive User Experience

Dynamic filtering, KPI cards, reset navigation buttons, and interactive visual elements were added to create a smooth and user-friendly analytical experience..

Performance Benefits Achieved

The optimization techniques resulted in:

- Faster dashboard loading
- Improved filter responsiveness
- Better visual performance
- Enhanced readability and user interaction
- Reduced data model complexity